



(Left to Right): Puneetha Reddy, Shabrin Firdose, Lokesh K.N., Madan Babu, Vybhav R., Venkatesh Gowda, Dr Sondur Lakshminipathi (sitting), Pradhan, Ranjith N., Valentino D'Souza, Jandran V., Devika Raghavendran and Srinivas Vinjamarra

The team

- Dr Sondur Lakshminipathi, founder and CTO, Mymo Wireless
- Madan Sondur, founding employee, director, product strategy
- Devika Raghavendran, founding employee, VP for technology
- Srinivas Vinjamarra, VP for digital signal processing - system on chip (DSP-SoC)
- Prashanth Nagarajan, VP for business development
- Venkataraman S., GM, products

Technical specifications

Frequency range: 70MHz-6GHz

Bandwidth: All LTE bandwidths (1.4, 3, 5, 10, 15, 20)

Duplex mode: Frequency division multiplexing (FDD) and time division multiplexing (TDD)

Power requirement: 15W

RF transmitting power: +26dBm

Speed: Uplink: 75Mbps; Downlink: 150Mbps

Range of operation: Up to 1km for a pico cell and between 30km and 100km for a macro cell

revenue generated from the customers increased, the fund requirements reduced. They moved on from technology licensing to customisation to royalty, based on particular requirements of global customers.

Developing the infrastructure of feared challenges of its own. The first challenge was in developing the LTE. But what turned out to be more difficult was proving it in real-time RF as a product. It was a high-technology product development that required capital-intensive infrastructure.

The next challenge was acquisition of expensive software and hardware tools/technology for validating the accuracy of the design and its implementation. So Mymo received support from customers with free access to their labs for test and vali-

dation, along with software and hardware tools. Also, using their in-house product, MW2000, reduced the development, validation and verification time by 100 folds, with lesser manpower.

The development roadmap was long and highly competitive with near monopolies like Qualcomm,

Huawei, Sequans and Altair that are established legacy product design houses. Venturing into such a high-tech product game required highly-talented human resources. The requirement then turned to a problem of retaining the talent, with competitive pay packages. A good balance on the pay and work culture proved right for them.

How unique it is

Offering LTE IP as a product turns out to be unique. There are competitors in the USA and Europe, but none, apart from big players, offer all three layers of the LTE protocol stack as a complete solution. Coming to their USR, MW2000, it is a first-of-its-kind LTE UE USR, Tier 4 SDR offering.

Mymo owns the technology be-

hind this central processing unit based modem and thus has the flexibility to modify the design to fit requirements. Owning the IP and physical product, along with the ability to adapt to different market needs, is what differentiates Mymo Wireless from its peers. The UE modems are currently being customised for niche markets like avionics, high-speed railways, defence and satellite communications based on LTE. In addition, Mymo offers the best business value with flexibility of business components suitable for the customers, like one-time royalty cost to them on their technology licensing.

The road ahead

Coming out of incubation from Indian Institute of Science in 2009, Mymo Wireless intends to go commercial with its full-blown commercial LTE IP and LTE products both in the local market and worldwide. Currently, the IP from Mymo is being used by customers in the USA including US defence, Europe including European civil aviation products, for Indian military in Military College of Telecommunication Engineering, Mhow, Indore and research organisations like Indian Institute of Technology, Hyderabad.

With the huge market space of the IoT/M2M, Mymo Wireless, along with collaboration with companies from the USA and Europe, would be tapping out one-of-a-kind ultra-low-power IoT/M2M LTE chip in the first quarter of 2016.

Mymo is also promising to release its own series of LTE chips for Indian and local markets ranging from the IoT/M2M, Cat1 to Cat4 for smartphones and other devices to cater to all LTE market requirements.

For all of the above, MWx000 is being used as proof-of-concept for pre-silicon IP validation by Tier 1 vendors like Verizon. The UE itself, as a product, is certified by Global Certification Forum. With LTE evolving, you can expect Mymo Wireless to aggressively follow suit and be the next Indian Qualcomm or Tier 1 company in this LTE domain worldwide. ●